GCBF: Grouped Cross-Band Fusion Network for Multispectral Scene Classification

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***Abstract*— Remote sensing scene classification is a crucial task for remote sensing image interpretation. Existing multispectral scene classification methods have overlooked the interrelationships between different spectral bands, which limits the mining of complementary information within the images. Addressing this issue, we propose a grouped cross-band fusion (GCBF) network for remote sensing multispectral scene classification to take full advantage of complementary information between various spectral bands. First, we separate the various bands of the given multispectral image into different groups to better capture the characteristics of each spectral band. Then, we use the existing UniFormer as a feature extractor to learn the representations of red, green, and blue (RGB) bands. For the spectral bands other than RGB, we propose a new network called multistage grouped spectral feature extraction (MGSFE) network to learn discriminative representations. We also draw inspiration from the band combination in the field of remote sensing and introduce a cross-band attention fusion (CBAF) module designed to adaptively merge features from both the RGB bands and other spectral bands. Extensive experiments on three widely used remote sensing multispectral scene classification datasets of BigEarthNet, SEN12MS, and EuroSAT demonstrate the superiority of our proposed method compared with several state-of-the-art (SOTA) methods.**

***Index Terms*— Cross-band attention, multispectral image, remote sensing scene classification.**

I. Introduction

REMOTE sensing scene classification is a key and fundamental task of remote sensing image interpretation. Specifically, it aims to label the given remote sensing images with correct semantic categories [1] (e.g., "Marine waters," "Industrial buildings," and "Arable land") by understanding the semantic information contained in the images. Scene classification can be applied in various fields, such as land use/cover investigations [2], natural disaster detection [3], urban planning [4], vegetation mapping [5], agriculture [6], [7], [8], and environment monitoring [9], [10]. With the rapid advancement of Earth observation technology, we can now obtain a vast amount of remote sensing images about the Earth's surface through satellites on a daily basis. The effective utilization of this massive image data for more meaningful Earth observation has gradually become a research hotspot of the remote sensing community. Consequently, remote sensing scene classification has gained increasing attention in recent years.

Currently, the vast majority of remote sensing scene classification datasets are created based on red, green, and blue (RGB) three-band images, such as the UC Merced land use dataset [11], the aerial image dataset [12], and so on. Although RGB images are the image type closest to human visual perception, for some highly similar scene categories, relying solely on RGB images is insufficient for accurate classification. This is because the texture features of some scenes in RGB bands are very similar, and RGB bands do not contain sufficient information to differentiate them. Due to the fact that various materials typically have different reflectivity for different wavelengths, and multispectral images have high spectral resolution and wide spectral range, the use of multispectral images can achieve more accurate recognition of scenes. For instance, an object may display a specific spectral reflectance pattern in the visible light range (such as RGB) and another pattern in the near-infrared range. These varied patterns reflect the objects' distinct biochemical or physical properties, such as water content, chlorophyll concentration, or surface temperature. From this, it can be seen that different bands with different data distributions can provide sufficient complementary information for scene classification tasks.

Recently, deep learning has demonstrated excellent capabilities in various fields, and many researchers have begun to explore the application of deep learning in remote sensing. Due to the relatively smaller number of remote sensing multispectral datasets compared to RGB datasets, research on remote sensing classification algorithms for RGB images dominates. However, due to the fact that multispectral images have more channels compared to RGB images, these RGB image-oriented methods cannot be directly used to cope with multispectral visual tasks. Wang et al. [13] apply deep neural networks (DNNs) designed for RGB images to multispectral

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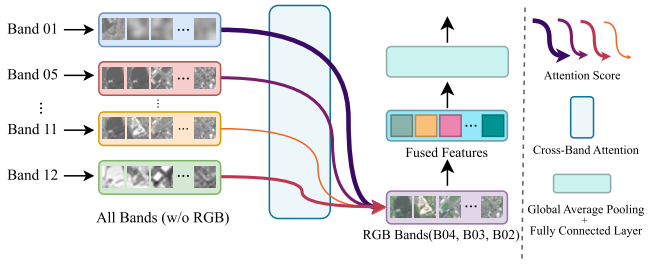


Fig. 1. Overview of GCBF. We group different bands and then use a cross-band attention mechanism to allow the network to adaptively combine these spectral bands.

images by changing the input layer of the network to adapt to the shape of multispectral images and then leverage the weights from the models pretrained on large-scale natural image datasets (e.g., ImageNet [14]). Although the method above can utilize the knowledge learned from natural RGB images, the substantial distribution difference between natural images and multispectral remote sensing images will limit further improvement in network performance. In [15], a 1×1 2-D convolutional filter is used to extract the spectral features. However, due to the relatively simple network structure, the extracted spectral features are not precise enough. Now, there are also some methods designed specifically for multispectral image classification. Unfortunately, these methods have not taken into account the interrelationships between different bands in multispectral images but simply group different bands based on their spatial resolution [16], [17], [18] or concatenate all bands directly [19], [20].

Unlike previous methods that overlooked the complementary information among spectral bands, we are inspired by the band combination in remote sensing and propose the grouped cross-band fusion (GCBF), aiming to achieve better multispectral scene classification performance by taking advantage of interband relationships, as shown in Fig. 1. Band combination refers to the allocation of invisible bands to RGB channels to generate a false color image, which is used to highlight the presence of certain materials, such as vegetation. Taking the Sentinel-2 multispectral image as an example, the combination of bands B08, B04, and B03 is more advantageous for researchers to analyze vegetation cover levels in different regions compared to other bands. In the proposed method, we consider the combination of RGB bands as one group, which, for Sentinel-2, is [B04, B03, B02]. Each band except the RGB is considered a separate group, which are [B01], [B05], [B06], and so on. Since the RGB bands have rich texture visual information, feature extraction methods for RGB images are well-established and effective. We utilize the existing UniFormer [21] as a feature extraction branch in GCBF to extract RGB bands' features. At the same time, we design the multistage grouped spectral feature extraction (MGSFE) network as another branch to process the bands other than RGB individually. The difference between UniFormer and our MGSFE is that the network structure of MGSFE is specifically designed for multiband data, which allows it to group feature extraction and feature enhancement for different bands. In the proposed cross-band attention fusion (CBAF) module, we treat RGB features as queries and features from other bands as keys and values. Through the cross-attention layer, the network can adaptively aggregate features from different spectral bands to model the dependencies between various band groups. The cross-attention mechanism used here can be seen as an adaptive band combination, which infers different attention maps for each input image, thereby achieving better classification performance than existing methods.

The main contributions are summarized as follows.

1) We review the limitations of existing deep learning methods for multispectral scene classification. Inspired by the band combination method, we attempt to explore the interrelationships between different bands in multispectral images to take full advantage of complementary information.
2) We group the spectral bands and propose the MGSFE branch network in GCBF to learn the distinct representations of each band group except for the RGB. We also propose the CBAF module to achieve an adaptive band combination by applying the cross-attention mechanism between varying band groups.
3) We evaluate the proposed method on three widely used multispectral scene classification datasets of BigEarthNet [22], SEN12MS [23], and EuroSAT [24]. The experimental results show that our method exhibits superior classification performance and achieves state-of-the-art (SOTA) performance across all evaluation metrics.

II. Related Work

A. Visible-Band Remote Sensing Scene Classification

Since visible remote sensing images are easy to acquire, there has been a surge in research on visible remote sensing scene classification in recent years. All the methods can be mainly categorized into traditional methods and deep learning-based methods.

The features used in traditional remote sensing scene classification methods can generally be divided into two types: handcrafted low-level visual features and mid-level visual features. Yang and Newsam [25] were the first to apply the scale-invariant feature transform (SIFT) [26] operator to scene classification, which can extract distinctive and invariant low-level visual features near key points. dos Santos et al. [27] explored the performance of color and texture descriptors in the feature representation for remote sensing scene classification and retrieval tasks. Risojević et al. [28] evaluated the feature description capabilities of Gabor and Gist descriptors on the aerial image classification dataset. Handcrafted features can only represent local features and do not contain semantic information. Some researchers have improved scene classification performance by encoding low-level features to obtain mid-level features, which can capture the semantic content of images. The most commonly used method for encoding handcrafted features is the bag-of-visual-words (BoVW). BoVW and its variants have demonstrated excellent classification performance [29], [30], [31].

With the outstanding performance of deep learning in many fields, an increasing number of researchers are exploring the

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application of deep learning in the field of scene classification. Cheng et al. [32] adopted two simple strategies to investigate the application of convolutional neural networks (CNNs) in scene classification. Xia et al. [33] proposed a dual-branch global–local attention network (DBGA-Net) to alleviate the problem of intraclass diversity and interclass similarity in remote sensing images. Lv et al. [34] simultaneously considered local structural information and the contribution of different channels, proposing a new vision transformer (ViT) model for scene classification. Dimitrovski et al. [35] conducted extensive experiments on the classification performance of ten different SOTA models on 22 scene classification datasets. For the multispectral datasets, they only computed the images in the RGB color space. Tang et al. [36] combined the advantages of CNN and transformer to propose a new efficient multiscale transformer and cross-level attention learning (EMTCAL), aiming to fully mine the information in remote sensing scenes. Wang et al. [37] explored the extraction and fusion of multiscale multilevel features from remote sensing images, achieving better scene classification results.

The visible remote sensing images contain only three spectral bands: RGB, which limits the further improvement of classification performance. In contrast, our method is based on multispectral images, allowing us to leverage more spectral information to achieve better classification results.

B. Multispectral Remote Sensing Scene Classification

Due to the fact that large-scale multispectral classification datasets have only been accessible in recent years and the current popularity of deep learning, existing multispectral remote sensing image scene classification methods focus more on deep learning methods.

Sumbul and Demİr [17] grouped multispectral images based on the varying spatial resolutions of different multispectral bands, and then, they employed a k-branch CNN and bidirectional long short-term memory (LSTM) networks for feature extraction. Chaudhuri et al. [18] designed a new representation learning network for remote sensing scene classification. Similar to [17], the spectral bands are also grouped based on their spatial resolutions. Some researchers have introduced the concept of knowledge distillation (KD) into multispectral scene classification, which allows for the transfer of knowledge across different modalities between teacher and student networks. Liu et al. [38] proposed the one-step cross-modal KD framework and two-step cross-modal KD framework with a teacher assistant (TA), transferring the knowledge from the teacher model pretrained on RGB images to the multispectral student model. Shin et al. [39] introduced a KD framework named multispectral-to-RGB KD (MS2RGB-KD). Opposite to the modality transfer direction in [38], the teacher of MS2RGB-KD is pretrained on the multispectral images, while the input to the student model is an RGB image. In [15], a novel bandwise multiscale (BWMS) network is proposed that can decouple the feature extraction process of spatial and spectral information in multispectral images. Furthermore, Papoutsis et al. [19] benchmarked 62 models on the BigEarthNet dataset. In experiments, all channels of the multispectral images are directly concatenated and input to the model without the dedicated design of the network structure according to the cross-band relationships.

The above multispectral image classification methods do not fully exploit the relationships between different spectral bands; they merely grouped or concatenated channels briefly. There are also some works that have designed models based on the interrelationships between different multispectral bands. Cong et al. [40] proposed SatMAE, which enhances the models' ability to extract multispectral features by grouping multispectral data and designing spectral positional encodings. Bountos et al. [41] introduced a foundation model named FoMo-Net, which is capable of handling various spectral bands. It includes trainable spectral embeddings, allowing the network to effectively capture the feature information unique to each spectral band. To explore the complementarity between different bands, we propose the GCBF, which groups spectral bands and uses a cross-attention mechanism to enable the network to learn more intrinsic relationships between different channels.

C. Attention Mechanism

The attention mechanism is now one of the most popular methods in the field of deep learning, inspired by the tendency of human biological systems to focus on the most important information rather than all of the information [42].

Mnih et al. [43] pioneered the combination of DNN and attention mechanism. Hu et al. [44] proposed a novel channel-attention architectural unit (i.e., SE block) to model the relationships between the channels. Woo et al. [45] proposed CBAM that can infer attention maps along spatial and channel dimensions, further improving the classification and detection performance of the network. Vaswani et al. [46] first introduced the self-attention mechanism and made a huge breakthrough in the field of natural language processing. Dosovitskiy et al. [47] proposed ViT, which is the first to apply the pure transformer architecture to computer vision tasks. The self-attention mechanism is designed to extract features from a single sequence, whereas cross-attention can merge two sequences belonging to different modalities. By using a cross-attention mechanism, we can better fuse data from diverse modalities. For instance, Stable Diffusion [48] utilizes cross-attention to merge image and text during image generation. In BLIP-2 [49], learned queries can interact with image features through the cross-attention mechanism.

The proposed CBAF module utilizes a cross-attention mechanism between different bands to facilitate interaction between different spectral bands, allowing the network to fully learn the interdependence between each channel.

III. Method

In this section, we first briefly describe existing frameworks for multispectral image processing and introduce the motivation and detailed framework of our proposed GCBF. Then, the three components of GCBF are introduced in sequence, including UniFormer used for RGB bands' feature extraction, MGSFE for feature extraction of bands other than RGB, and CBAF for the feature fusion of different band groups.

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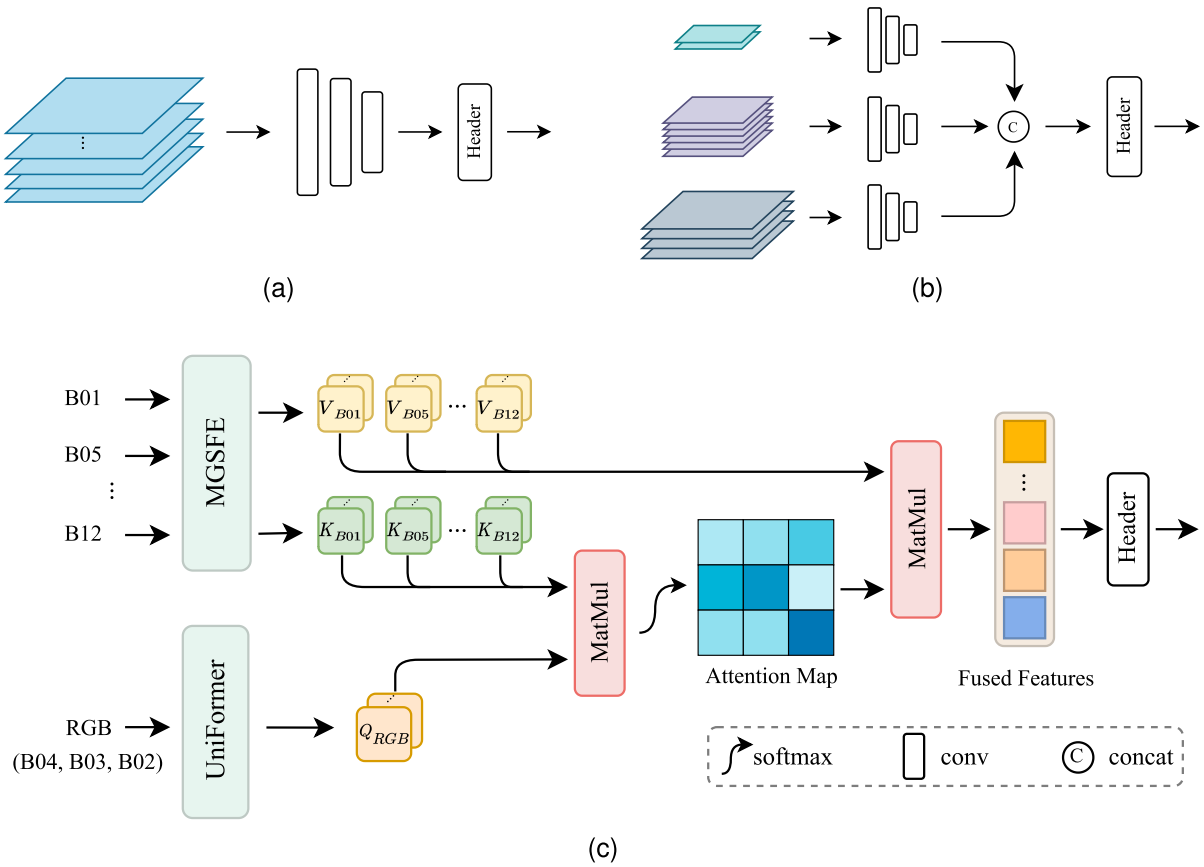


Fig. 2. Motivation of our framework. (a) This method upsamples the images of all bands to the same resolution, then concatenates them, and inputs them into the model. (b) Spectral bands with the same resolution are divided into the same group, and all groups are fed into a multibranch CNN. Finally, the features extracted by each branch are concatenated and input into the classification header. (c) Our GCBF is proposed to thoroughly exploit the interrelationships between different bands through cross-band attention, thereby learning better representations of multispectral images.

A. Motivation and Framework

Multispectral images have more channels and contain a greater amount of spectral information compared to RGB images. There are currently two main processing frameworks for multispectral images. The first is to upsample images from different spectral bands to the same spatial resolution and then concatenate them along the channel dimension before feeding them into the model, as shown in Fig. 2(a). As illustrated in Fig. 2(b), another framework is to group images according to their different spatial resolutions and then input each group into the corresponding network. Finally, the extracted features from each branch are concatenated and input into the classification headers to obtain the predictions.

The two multispectral image processing frameworks mentioned above do not take into account the interrelationships between spectral bands. Based on the limitations of existing methods and inspired by band combination, we propose GCBF to address this issue, as shown in Fig. 2(c). In GCBF, we consider the RGB bands as one group and each of the remaining bands as a separate group. For images in each group, we first split them into patches. Then, we utilize the existing UniFormer to extract features from the RGB group. We design MGSFE to work as the feature extractor for the remaining band groups. MGSFE can enhance the representations of each band through multistage groupwise feature extraction and fusion. Finally, the features extracted by UniFormer and MGSFE interact within the cross-band attention layer of CBAF. In summary, our GCBF mainly consists of three parts. Two feature extraction branch networks, UniFormer and MGSFE, extract features from different band groups, respectively, and a CBAF module is used to fuse features from different bands to complete the final classification. The detailed introduction of each key component is given as follows.

B. RGB Bands' Feature Extraction

In the processing of multispectral images, we group all the spectral bands, noting that the three bands of RGB are treated as one group. The decision to combine the RGB bands into one group is based on two considerations. First, RGB data are close to the human visual perception system, containing rich visual information that humans can perceive. Consequently, combining the RGB bands into one group for processing aligns with human visual habits in understanding remote sensing scenes. Second, many computer vision algorithms are designed for RGB images, which means that the current network architectures for RGB image feature extraction are very mature and robust. Combining RGB bands can maximize the utilization of the information contained in these three bands.

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In our proposed GCBF, we use UniFormer as an RGB feature extraction branch network to obtain the representations of RGB bands. UniFormer is a powerful backbone proposed in [21] to address the large local redundancy and complex long-range dependence of visual information (e.g., images and videos). It unifies convolution and self-attention mechanisms, overcoming the limited receptive field of convolution and the excessive focus of self-attention on global dependence in shallow layers. According to different parameter sizes, there are three types of UniFormer: UniFormer-Small, UniFormer-Base, and UniFormer-Large. To find a better balance between performance and computation cost, we opt for the UniFormer-Small here. For more details about UniFormer architecture, please refer to [21].

C. Multistage Grouped Spectral Feature Extraction

The primary distinction between multispectral images and RGB images is that multispectral images have more spectral bands and richer information. Moreover, since different spectral bands capture distinct reflection characteristics of scenes, each band contains unique information. To explore the interrelationships between different bands more effectively, it is necessary to consider how to enhance the extraction of information from each band other than RGB bands. Here, we propose MGSFE as a feature extraction branch of GCBF, with the aim of multistage feature extraction and enhancement from different spectral bands other than RGB bands.

The overall architecture of the MGSFE is presented in Fig. 3(a). It consists of a total of four stages, and all stages are composed of alternating stacking of patch embedding layers and MGSFE blocks. The numbers of MGSFE blocks in the first, second, third, and fourth stages are 3, 3, 9, and 3, respectively.

Due to the redundancy of information in images, the downsampling operation is usually performed first in CNNs and ViTs. In ResNet, the downsampling layer typically consists of a 7×7 convolution layer with stride 2 and a max pooling layer. In ViTs, it is common to use the "patchify" layer to split the image into patches for downsampling [50]. Because UniFormer uses the "patchify" strategy for downsampling the RGB bands, to keep the feature space alignment of all bands, in MGSFE, we use the same downsampling strategy with UniFormer. It should be noted that in the first patch embedding layer, we adopt the normal "patchify" strategy (setting the number of groups is 1) to enable the initial interaction of information from different spectral bands. In the deep layers, in order to preserve the learned features of each band group from being influenced by other groups, we use grouped patch embedding. In MGSFE, the normal patch embedding strategy is implemented the same way as in ViT, while grouped patch embedding is achieved using groupwise convolution (GWConv). It is important to note that the grouping is applied at the spectral band granularity, with each spectral band assigned a dedicated convolution operation.

In order to fully learn the distinctive representations within each band, we primarily utilize GWConv, pointwise convolution (PWConv), layer normalization, and a nonlinear activation to form each block, as depicted in Fig. 3(b). Given the input feature map $X_i \in \mathbb{R}^{H\times W\times(N_B\cdot C)}$, where N_B denotes the number of bands input in the MGSFE, note that it does not include RGB bands. C denotes the channel number of each group. The MGSFE block can be formulated as follows:

$$X = \mathrm{GWConv}(X_i) \tag{1}$$

$$Y = \mathrm{PWConv}(\mathrm{Norm}(X)) \tag{2}$$

$$Z = \mathrm{PWConv}(\mathrm{GELU}(Y)) + X_i \tag{3}$$

where X and Y are the outputs of GWConv and the first PWConv, and Z is the output of the current MGSFE block.

We treat each band outside of RGB bands as a separate group and extract features for each group separately because GWConv can process different dimensions of the input feature map separately through multiple groups of convolution kernels. Therefore, we use GWConv to achieve feature extraction for different bands, where the number of groups of convolutional kernels is equal to the number of bands in the input. Ding et al. [51] reveal that larger kernel CNNs have much larger effective receptive fields and higher shape bias, so we follow the work of [50], using GWConv with a convolution kernel size of 7×7. Subsequently, after layer normalization, the output of GWConv X is sequentially fed into two PWConv layers with a GELU activation in between to provide nonlinearity. In addition, a residual connection is applied at the beginning and end of the block. In the MGSFE block, two PWConv layers are used for simple interaction of information between spectral bands and feature enhancement. The more comprehensive feature interaction will be achieved in CBAF.

D. Cross-Band Attention Fusion

Through UniFormer and MGSFE, we have obtained features for each spectral band group. Next, it is crucial to determine how to fully leverage the complementary information and dependencies between features from different band groups. In our method, we design the CBAF to address this challenge. Band combination refers to the allocation of invisible bands to RGB channels to generate a false color image, which can be seen as an operation of modeling the interrelationships between different bands. In deep learning, the attention mechanism is very effective in modeling the dependencies between input data. As we know, the self-attention mechanism can only handle unimodal input sequences. Hence, we apply the cross-attention mechanism in CBAF to better merge the input sequences from different band groups. In addition, if we consider band combination as a weighted sum operation, where the selected bands have higher weights and the unselected bands have lower importance, then the weights in traditional band combinations are static. However, the cross-attention mechanism can generate dynamic weights, as the calculations of weights vary with different input images [52]. Consequently, our proposed CBAF based on cross-attention can achieve dynamic band combinations for different input images and has a more robust performance. Furthermore, for the traditional band combination method, only some typical scenes have band combination schemes (e.g., water and vegetation),

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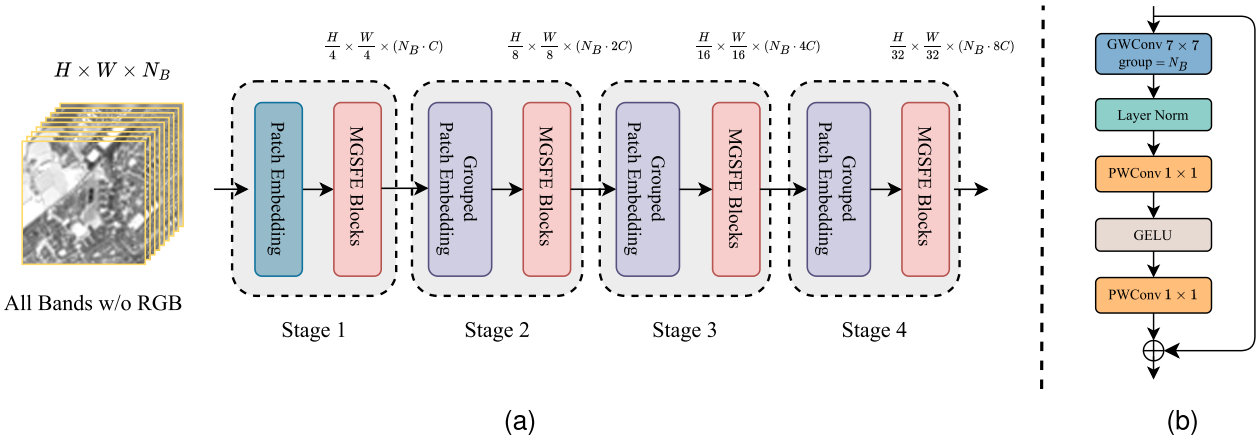


Fig. 3. (a) Overall network architecture of our proposed MGSFE, which consists of four stages. Each stage is stacked by normal/grouped patch embedding layers and MGSFE blocks. The input size is $H \times W \times N_B$, where N_B is the number of spectral bands. The size of the feature maps output by each stage is illustrated at the top. Note that the multispectral images input to MGSFE do not include the RGB bands. (b) Detailed structure of the MGSFE block. The main components are GWConv, PWConv, layer normalization, and GELU activation.

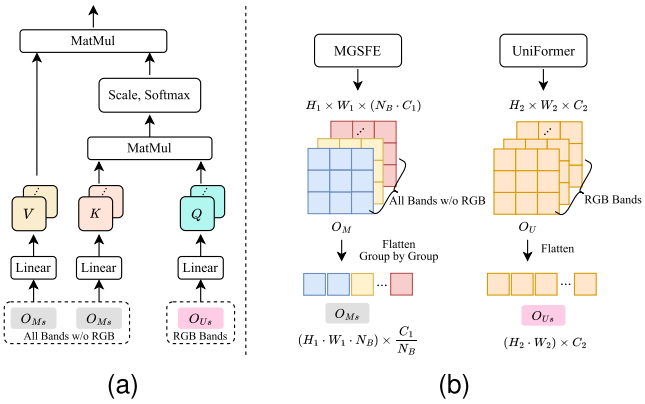


Fig. 4. (a) Illustration of CBAF architecture. O_{Us} and O_{Ms} are the feature token sequences from RGB bands and other band groups, respectively. (b) Detailed process of flattening O_M and O_U output by MGSFE and UniFormer into O_{Ms} and O_{Us}.

TABLE I
ALL 13 SPECTRAL BANDS OF THE SENTINEL-2

Band	Resolution. (m)	Central Wavelength (nm)	Bandwidth (nm)	Description
B01	60	442.7	20	Coastal aerosol
B02	10	492.7	65	Blue
B03	10	559.8	35	Green
B04	10	664.6	30	Red
B05	20	704.1	14	Vegetation red edge
B06	20	740.5	14	Vegetation red edge
B07	20	782.8	19	Vegetation red edge
B08	10	832.8	105	NIR
B8A	20	864.7	21	Narrow NIR
B09	60	945.1	19	Water vapour
B10	60	1373.5	29	SWIR - Cirrus
B11	20	1613.7	90	SWIR
B12	20	2202.4	174	SWIR

while most ground objects do not have suitable band combination schemes. In contrast, our GCBF can learn corresponding band combination schemes for all classes in the dataset during the training process, resulting in improved versatility.

To apply the cross-band attention operation, we first need to reshape the feature maps $O_M \in \mathbb{R}^{H_1 \times W_1 \times (N_B \cdot C_1)}$ and $O_U \in \mathbb{R}^{H_2 \times W_2 \times C_2}$ output by MGSFE and UniFormer into a sequence of flattened 2-D tokens. In Fig. 4(b), we show the detailed process of flattening O_M and O_U into token sequence $O_{\mathrm{Ms}} \in \mathbb{R}^{N_1 \times C_1}$ and $O_{\mathrm{Us}} \in \mathbb{R}^{N_2 \times C_2}$, where $C1 = C_1/N_B$ is the channel dimension of each band group, and $N_1 = H_1 \times W_1 \times N_B$ and $N_2 = H_2 \times W_2$ are the length of tokens sequence. Notably, O_M is flattened group by group. Then, we input the obtained feature sequences of each band into CBAF for adaptive fusion, as shown in Fig. 4(a). Queries Q, Keys K, and Values V embeddings are three key elements in the cross-attention mechanism, which can be obtained as follows:

$$Q = O_{\mathrm{Us}} W_q, \quad K = O_{\mathrm{Ms}} W_k, \quad V = O_{\mathrm{Ms}} W_v \tag{4}$$

where $W_q \in \mathbb{R}^{C_2 \times C_2}$, $W_k \in \mathbb{R}^{C_1 \times C_2}$, and $W_v \in \mathbb{R}^{C_1 \times C_2}$ are the projection matrices corresponding to Q, K, and V. The learning of dependencies between different bands is based on a pairwise comparison between the queries from RGB bands and keys from other bands. In detail, the comparison between tokens is achieved through the dot-product operation. At the same time, in order to alleviate the problem of possible gradient vanishing, the results of the dot product are scaled with $1/(C_2)^{1/2}$. Then, the softmax function is applied to the scaled result to obtain the attention score. Finally, the enhanced band features are generated by using the attention scores as weights to perform a weighted sum operation on values. In summary, the workflow of CBAF can be formulated as follows:

$$O_C = \text{Cross-Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{C_2}}\right)V \tag{5}$$

where $O_C \in \mathbb{R}^{N_2 \times C_2}$ is the output of CBAF. Then, we input O_C into the classification header to obtain the final scene classification results.

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IV. Experiment

A. Datasets

Extensive experiments are conducted to evaluate the performance of the proposed GCBF on the widely used remote sensing multispectral datasets of BigEarthNet [22], SEN12MS [23], and EuroSAT [24].

The BigEarthNet is a large-scale multispectral multilabel land-cover image classification dataset, which consists of 590 326 multispectral images of 19 classes in total. All image patches are from Sentinel-2 and comprise 13 spectral bands. The band B10 is excluded because it does not contain surface information. For the bands at 10, 20, and 60 m, the image sizes are 120 × 120, 60 × 60, and 20 × 20. In addition, 70 987 images heavily covered by cloud, cloud shadow, and seasonal snow were discarded following the setting in [22]. We follow the instructions of [13] for the train, val, and test splits on BigEarthNet.

The SEN12MS dataset is also a multilabel, multispectral remote sensing image scene classification dataset. This dataset comprises a total of ten categories and includes 180 662 images. All image patches are from Sentinel-2 and comprise ten spectral bands. For the division of the training set, the validation set, and the test set, we follow the official partition settings.

The EuroSAT is a multispectral single-label scene classification dataset and contains 27 000 multispectral images of ten classes. These images are also from Sentinel-2 and have 13 spectral bands with a size of 64 × 64. To remain consistent with the BigEarthNet dataset, the tenth band of EuroSAT is also removed. Considering that the number of images varies for each class, in order to ensure the reliability of the evaluation results of our method, we divide 540 samples from each category into the test set, which accounts for 20% of all images. The remaining images are used for training and validation.

To better illustrate the characteristics of the images in BigEarthNet, SEN12MS, and EuroSAT, we provide the spectral band information of Sentinel-2 satellites in Table I.

B. Implementation Details

All experiments are conducted on an NVIDIA Tesla V100 GPU using PyTorch. In fact, current research on multispectral scene classification is scattered and scarce. For a comprehensive comparison, in addition to the SOTA multispectral classification method WRN-B4-EC [19] on BigEarthNet and BWMS [15], we also select several representative and powerful models for the following comparison experiments. They are ResNet-152 [53], ViT-L [47], Swin-B [54], EMTCAL [36], GCSANet [55], and EFPN-DSE-TDFF [37].

In our experiments, both our proposed GCBF and the compared models are trained from scratch, which means that the weights of the models are initialized randomly. Due to differences in dataset sizes, we have different experimental settings on BigEarthNet, SEN12MS, and EuroSAT for some parameters. Our model is trained using AdamW with $\beta_1 = 0.9$, $\beta_2 = 0.999$, and a momentum of 0.9 for both datasets. We also use a cosine learning rate schedule to adjust the learning rate during training. For BigEarthNet/EuroSAT, the batch size, the weight decay, total epochs, and the learning rate are set to 64/64, 0.01/0.07, 100/300, and 0.0005/0.0006, respectively. In addition, the first ten epochs are utilized for linear warm-up for EuroSAT. For EMTCAL, GCSANet, BWMS, and EFPN-DSE-TDFF, we use the same parameters as the original paper for training, including learning rate and so on, and the total epochs are set to 100 and 300 for BigEarthNet and EuroSAT datasets, respectively. The training parameter settings for the SEN12MS dataset are consistent with those of BigEarthNet.

Taking into account the differences in model architecture and for a fair comparison, we train different models with appropriate parameters and optimizers. In our experiments, we find that training ResNet, ViT, and Swin transformers from scratch using the AdamW optimizer is challenging to converge. To obtain comparable and fair results, an SGD optimizer with a momentum of 0.9 is used to train these models. The epochs for these models trained on BigEarthNet, SEN12MS, and Eurosat are 100, 100, and 300, respectively. Due to the limitations of computational resources, we set different batch sizes for different models. For ResNet, the batch size is set to 128 for both datasets. ViT and Swin transformers have batch sizes of 32, 32, and 48 on BigEarthNet, SEN12MS, and EuroSAT.

In addition, to ensure the reliability of the experimental results, we conducted three experiments on each dataset with different random seeds. It should be noted that since WRN-B4-ECA [19] has not been open-sourced, the results on BigEarthNet are cited from the original paper.

C. Evaluation Metrics

To comprehensively evaluate the classification performance of each method, we use four evaluation metrics for EuroSAT, which are accuracy, precision, recall, and $F1$-score. Given that the performance evaluation of multilabel classification task is more complicated than single-label classification task [19], we report mean average precision (mAP), micro-precision/macroprecision (denoted as "OP"/"CP"), microrecall/macrorecall ("OR"/"CR"), and micro-/macro-$F1$-score ("OF1"/"CF1") for performance comparison on BigEarthNet and SEN12MS.

The accuracy can be formulated as follows:

$$\mathrm{Acc} = \frac{\mathrm{TP} + \mathrm{TN}}{\mathrm{TP} + \mathrm{TN} + \mathrm{FP} + \mathrm{FN}} \tag{6}$$

where TP and TN represent the number of samples correctly classified as positive and negative, respectively. FP and FN are the number of samples incorrectly classified as positive and negative, respectively. The definitions of precision P, recall R, and $F1$-score $F1$ are given as follows:

$$P = \frac{\mathrm{TP}}{\mathrm{TP} + \mathrm{FP}} R = \frac{\mathrm{TP}}{\mathrm{TP} + \mathrm{FN}} F1 = \frac{2 \times P \times R}{P + R}. \tag{7}$$

The mAP can be formulated as follows:

$$\mathrm{mAP} = \frac{1}{C} \sum_{i}^{C} \mathrm{AP}_i \tag{8}$$

$$\mathrm{AP} = \sum_{n} (R_n - R_{n-1}) P_n \tag{9}$$

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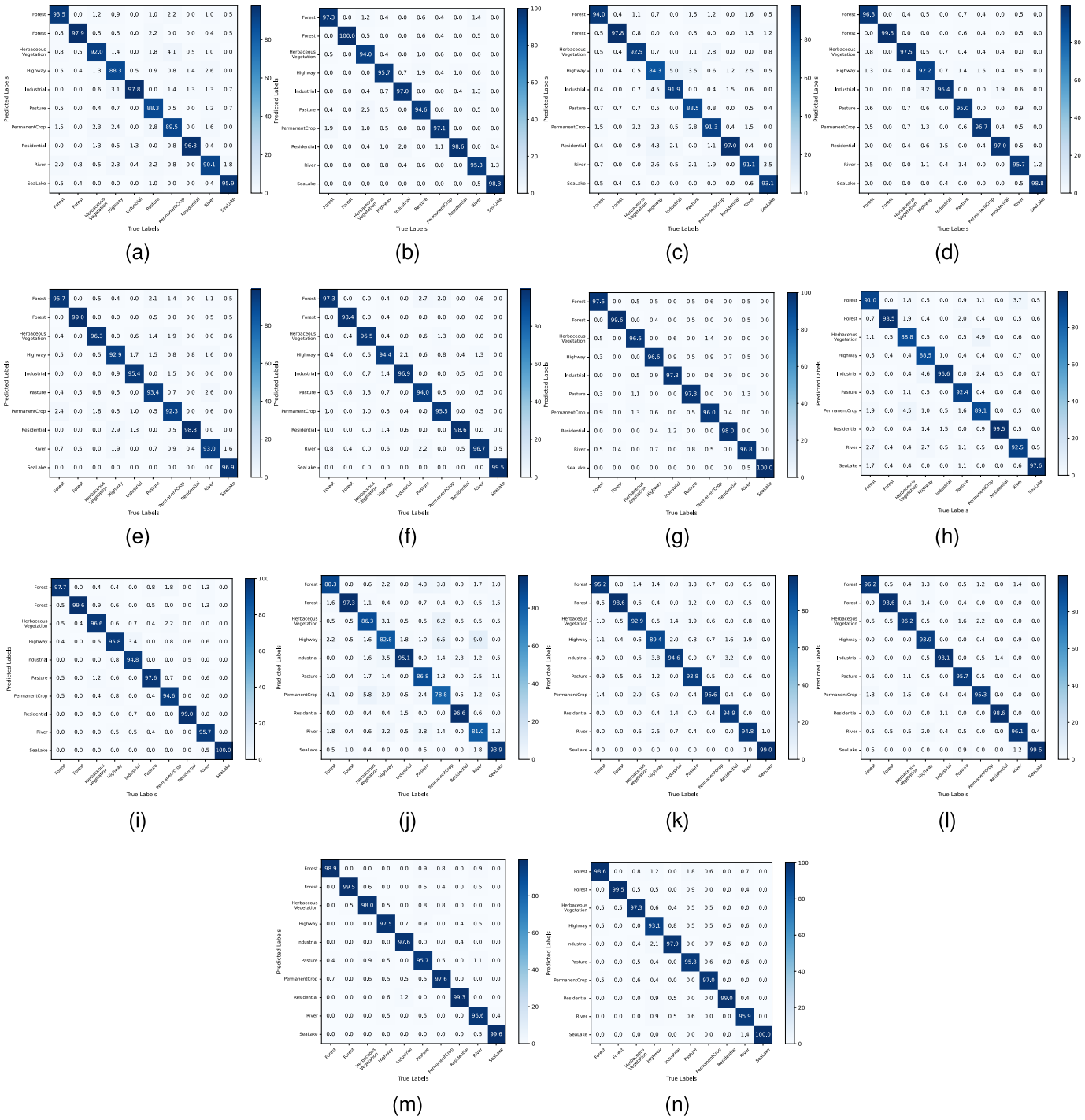


Fig. 5. Confusion matrices of different methods on EuroSAT. (a) ResNet-152$_{\text{RGB}}$. (b) ResNet-152$_{\text{ALL}}$. (c) ViT-L$_{\text{RGB}}$. (d) ViT-L$_{\text{ALL}}$. (e) Swin-B$_{\text{RGB}}$. (f) Swin-B$_{\text{ALL}}$. (g) EMTCAL$_{\text{RGB}}$. (h) EMTCAL$_{\text{ALL}}$. (i) GCSANet$_{\text{RGB}}$. (j) GCSANet$_{\text{ALL}}$. (k) EFPN-DSE-TDFF$_{\text{RGB}}$. (l) EFPN-DSE-TDFF$_{\text{ALL}}$. (m) BWMS. (n) GCBF (ours).

where C is the number of classes, AP_i denotes the average precision of class i, and R_n and P_n are the recall and precision at the nth threshold.

Following [56], the macroprecision/microprecision, macrorecall/microrecall, and macro-/micro-$F1$-score are defined as follows:

$$P_{\text{micro}} = \frac{\sum_{c=1}^{C} N_c^{\text{correct}}}{\sum_{c=1}^{C} N_c^{\text{predict}}} \quad P_{\text{macro}} = \frac{1}{C}\sum_{c=1}^{C} \frac{N_c^{\text{correct}}}{N_c^{\text{predict}}} \tag{10}$$

$$R_{\text{micro}} = \frac{\sum_{c=1}^{C} N_c^{\text{correct}}}{\sum_{c=1}^{C} N_c^{\text{gt}}} \quad R_{\text{macro}} = \frac{1}{C}\sum_{c=1}^{C} \frac{N_c^{\text{correct}}}{N_c^{\text{gt}}} \tag{11}$$

$$F1_{\text{micro}} = \frac{2 \times P_{\text{micro}} \times R_{\text{micro}}}{P_{\text{micro}} + R_{\text{micro}}} \quad F1_{\text{macro}} = \frac{1}{C}\sum_{c=1}^{C} F1_c \tag{12}$$

where N_c^{correct} is the number of correctly classified images for the cth label, N_c^{predict} denotes the number of predicted images for the cth label, N_c^{gt} represents the number of ground-truth

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TABLE II

QUANTITATIVE RESULTS OF OUR METHOD AND OTHER METHODS IN TERMS OF MAP, OP, OR, OF1, CP, CR, AND CF1 ON BIGEARTHNET WITH DIFFERENT RANDOM SEEDS. THE BEST RESULTS ARE MARKED IN BOLD

Method	mAP	OP	OR	OF1	CP	CR	CF1
WRN-B4-ECA	-	82.4	75.5	78.8	-	-	-
ResNet-152$_{RGB}$	82.23 ± 0.46	79.50 ± 1.35	64.70 ± 1.22	71.33 ± 0.49	68.17 ± 3.20	51.53 ± 1.21	57.10 ± 0.50
ResNet-152$_{ALL}$	88.03 ± 0.06	83.03 ± 0.06	73.03 ± 0.12	77.50 ± 0.36	74.77 ± 2.66	61.30 ± 0.98	66.13 ± 1.77
ViT-L$_{RGB}$	85.80 ± 0.36	80.27 ± 0.86	71.30 ± 1.40	75.47 ± 0.50	67.70 ± 0.36	59.47 ± 0.85	62.07 ± 0.51
ViT-L$_{ALL}$	88.93 ± 0.40	80.83 ± 1.10	77.57 ± 0.32	79.13 ± 0.67	72.03 ± 0.85	65.77 ± 0.65	67.83 ± 0.51
Swin-B$_{RGB}$	87.30 ± 0.30	81.37 ± 0.38	73.67 ± 0.12	77.30 ± 0.17	70.33 ± 0.32	60.43 ± 0.31	63.60 ± 0.46
Swin-B$_{ALL}$	89.37 ± 0.32	82.80 ± 0.44	76.40 ± 0.50	79.50 ± 0.35	73.77 ± 2.06	64.90 ± 0.36	67.93 ± 1.21
EMTCAL$_{RGB}$	85.73 ± 0.21	80.37 ± 0.64	71.27 ± 0.95	75.53 ± 0.32	74.10 ± 1.18	60.27 ± 1.02	65.37 ± 0.55
EMTCAL$_{ALL}$	87.27 ± 0.64	80.50 ± 0.53	75.10 ± 2.15	77.70 ± 1.11	74.83 ± 0.97	66.73 ± 2.87	69.73 ± 2.12
GCSANet$_{RGB}$	87.13 ± 0.42	81.70 ± 0.30	72.53 ± 0.55	76.87 ± 0.38	74.03 ± 0.61	60.50 ± 0.61	65.40 ± 0.66
GCSANet$_{ALL}$	89.03 ± 0.35	83.30 ± 0.20	74.97 ± 1.14	78.90 ± 0.56	76.43 ± 1.10	64.53 ± 1.40	69.00 ± 1.21
EFPN-DSE-TDFF$_{RGB}$	87.80 ± 0.30	81.87 ± 0.21	74.20 ± 0.20	77.87 ± 0.21	74.90 ± 0.36	62.70 ± 0.79	67.20 ± 0.70
EFPN-DSE-TDFF$_{ALL}$	88.40 ± 0.46	82.03 ± 0.21	75.30 ± 1.22	78.50 ± 0.62	75.23 ± 1.42	64.33 ± 3.69	68.30 ± 3.05
BWMS	87.77 ± 0.15	82.70 ± 0.30	72.80 ± 0.26	77.43 ± 0.21	74.03 ± 0.81	59.73 ± 0.58	64.80 ± 0.46
GCBF(ours)	**89.90 ± 0.20**	**84.77 ± 0.86**	**80.51 ± 0.70**	**82.57 ± 0.72**	**80.03 ± 2.12**	**72.33 ± 1.39**	**75.13 ± 1.77**

TABLE III

QUANTITATIVE RESULTS OF OUR METHOD AND OTHER METHODS IN TERMS OF MAP, OP, OR, OF1, CP, CR, AND CF1 ON SEN12MS WITH DIFFERENT RANDOM SEEDS. THE BEST RESULTS ARE MARKED IN BOLD

Method	mAP	OP	OR	OF1	CP	CR	CF1
ResNet-152$_{RGB}$	57.80 ± 2.25	57.63 ± 1.91	51.97 ± 2.56	54.60 ± 2.25	42.50 ± 2.03	38.13 ± 1.51	38.00 ± 1.87
ResNet-152$_{ALL}$	70.43 ± 3.16	67.70 ± 1.04	64.57 ± 3.49	65.97 ± 2.16	57.00 ± 1.81	46.40 ± 1.14	48.83 ± 1.22
ViT-L$_{RGB}$	63.43 ± 0.65	63.64 ± 1.96	56.90 ± 1.90	60.07 ± 0.86	46.90 ± 1.81	41.80 ± 1.71	42.73 ± 0.92
ViT-L$_{ALL}$	75.37 ± 1.02	67.33 ± 0.67	67.30 ± 0.60	67.27 ± 0.40	55.50 ± 1.00	51.37 ± 1.85	52.73 ± 0.81
Swin-B$_{RGB}$	73.77 ± 0.81	68.37 ± 0.55	65.57 ± 0.91	66.93 ± 0.51	53.50 ± 2.00	48.67 ± 0.95	49.70 ± 0.44
Swin-B$_{ALL}$	79.67 ± 0.81	72.70 ± 2.08	71.17 ± 1.53	71.90 ± 1.30	59.10 ± 1.56	53.20 ± 1.08	55.33 ± 0.21
EMTCAL$_{RGB}$	58.07 ± 0.85	58.77 ± 1.17	53.20 ± 0.89	55.80 ± 0.98	42.13 ± 0.60	41.10 ± 0.95	39.67 ± 0.84
EMTCAL$_{ALL}$	75.13 ± 0.60	68.33 ± 2.06	70.10 ± 1.22	69.20 ± 0.80	57.37 ± 2.27	52.73 ± 1.27	53.80 ± 0.70
GCSANet$_{RGB}$	68.30 ± 1.39	65.57 ± 1.15	56.90 ± 2.17	60.90 ± 0.98	49.73 ± 1.46	41.57 ± 2.40	42.97 ± 1.91
GCSANet$_{ALL}$	70.43 ± 1.15	69.47 ± 1.36	62.60 ± 1.47	65.83 ± 1.37	57.77 ± 1.91	46.20 ± 1.73	50.30 ± 1.73
EFPN-DSE-TDFF$_{RGB}$	59.27 ± 3.10	59.53 ± 1.42	55.77 ± 2.89	57.57 ± 1.76	43.77 ± 3.80	40.40 ± 5.15	40.37 ± 3.78
EFPN-DSE-TDFF$_{ALL}$	73.13 ± 0.71	66.77 ± 1.88	65.83 ± 1.25	66.27 ± 1.10	54.77 ± 1.31	50.07 ± 1.04	50.73 ± 0.90
BWMS	69.40 ± 1.25	64.00 ± 0.82	64.87 ± 1.36	64.40 ± 0.70	54.23 ± 1.26	46.03 ± 3.40	47.17 ± 2.35
GCBF(ours)	**81.77 ± 1.03**	**74.13 ± 1.53**	**72.87 ± 1.72**	**73.43 ± 0.21**	**60.03 ± 0.65**	**54.00 ± 1.06**	**55.40 ± 0.53**

images for the cth label, and $F1_c$ is the $F1$-score for the cth label. Apart from mAP, other metrics require setting a threshold during the prediction process, which we set at 0.5.

D. Comparison With the SOTA Methods

We compare the proposed GCBF with several SOTA methods, including WRN-B4-ECA [19], ResNet [53], ViT [47], Swin transformer [54], EMTCAL [36], GCSANet [55], EFPN-DSE-TDFF [37], and BWMS [15].

1) BigEarthNet: In Table II, we show the results of our GCBF and other SOTA methods on BigEarthNet, where the subscripts "$_{\text{RGB}}$" and "$_{\text{ALL}}$" mean using only RGB bands and all bands as input, respectively. BWMS is a method designed for multispectral images, utilizing all bands; hence, no subscript is annotated. All best results are marked in bold, and it is clear that the proposed GCBF outperforms all other methods in all metrics. In addition, comparing the methods using all spectral bands with the methods using only RGB bands, it can be seen that using more bands can indeed improve the classification performance, and our method can better harness the complementary information from all bands to achieve the best results.

2) SEN12MS: In Table III, we show the results of our GCBF and other SOTA methods on SEN12MS, where the subscripts "$_{\text{RGB}}$" and "$_{\text{ALL}}$" mean using only RGB bands and all bands as input, respectively. As shown in the table, all methods exhibit a decline in metrics on this dataset compared to the BigEarthNet dataset, indicating that this dataset presents a greater challenge. However, our method still achieves the best results across all metrics, demonstrating the effectiveness of GCBF in processing multispectral remote sensing images.

TABLE IV

QUANTITATIVE RESULTS OF OUR METHOD AND OTHER METHODS IN TERMS OF ACC, P, R, AND $F1$ ON EUROSAT WITH DIFFERENT RANDOM SEEDS. THE BEST RESULTS ARE MARKED IN BOLD

Method	Acc	P	R	F1
ResNet-152$_{RGB}$	93.70 ± 0.56	93.70 ± 0.62	93.63 ± 0.64	93.57 ± 0.71
ResNet-152$_{ALL}$	97.33 ± 0.50	97.33 ± 0.57	97.30 ± 0.56	97.20 ± 0.56
ViT-L$_{RGB}$	93.80 ± 0.92	93.57 ± 0.76	93.50 ± 0.72	93.13 ± 0.67
ViT-L$_{ALL}$	97.80 ± 0.17	97.73 ± 0.15	97.76 ± 0.15	97.47 ± 0.12
Swin-B$_{RGB}$	96.27 ± 0.23	96.20 ± 0.30	96.10 ± 0.30	95.60 ± 0.46
Swin-B$_{ALL}$	98.20 ± 0.10	98.13 ± 0.06	98.20 ± 0.10	97.93 ± 0.06
EMTCAL$_{RGB}$	94.83 ± 0.61	94.87 ± 0.67	94.80 ± 0.66	94.77 ± 0.67
EMTCAL$_{ALL}$	97.93 ± 0.21	97.90 ± 0.2	97.9 ± 0.20	97.83 ± 0.23
GCSANet$_{RGB}$	89.87 ± 0.32	89.83 ± 0.38	89.83 ± 0.29	89.67 ± 0.58
GCSANet$_{ALL}$	96.13 ± 0.06	96.17 ± 0.12	96.13 ± 0.06	96.07 ± 0.06
EFPN-DSE-TDFF$_{RGB}$	96.27 ± 1.44	96.40 ± 1.32	96.33 ± 1.38	96.20 ± 1.49
EFPN-DSE-TDFF$_{ALL}$	98.17 ± 0.15	98.20 ± 0.17	98.13 ± 0.15	98.13 ± 0.15
BWMS	97.97 ± 0.40	97.97 ± 0.40	97.90 ± 0.46	97.87 ± 0.40
GCBF(ours)	**99.00 ± 0.26**	**99.03 ± 0.31**	**98.97 ± 0.25**	**98.93 ± 0.31**

3) EuroSAT: To further verify the generalization of our method on different datasets, we report the results of all

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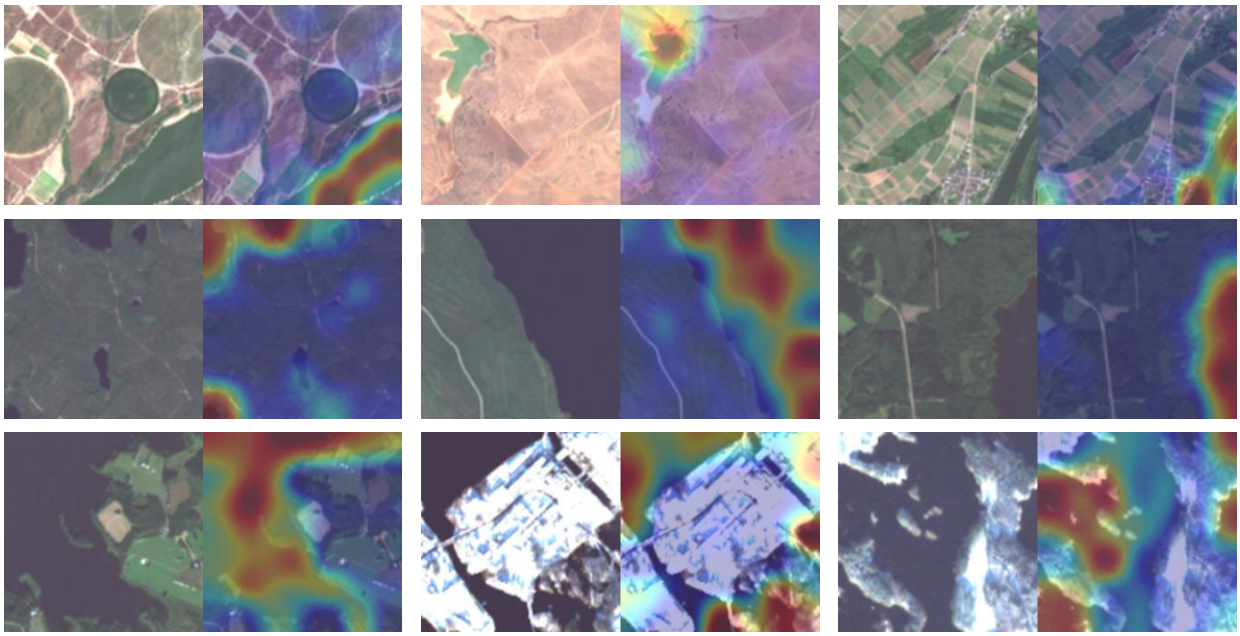


Fig. 6. Class activation maps for "water" in band B01 of GCBF. It can be seen that band B01 is sensitive to water, and our method can effectively focus on the water in the given images.

methods on EuroSAT in Table IV. As we can see from Table IV, our method still achieves the best performance on EuroSAT. Specifically, it achieves the Acc, P, R, and $F1$ of 98.7%, 98.7%, 98.7%, and 98.6%.

4) Visualization: In Fig. 5, we show the corresponding confusion matrices of different methods on EuroSAT. The number in the i-th row and j-th column represents the probability the j-th class is predicted as the i-th class. From the elements on the diagonal of the matrix, it can be seen that our method performs overall better than other methods. In order to verify the effectiveness of our method on representation learning for various spectral bands, we visualize the regions of interest for the bands B01 in GCBF, as shown in Fig. 6. The red area in the figure represents the areas that our model pays more attention to, while the blue areas indicate relatively less important areas compared to the red areas. In Fig. 6, it can be seen that our model has learned that band B01 is sensitive to water, which can effectively identify the water bodies and even narrow and elongated rivers in the given images.

To further validate that GCBF has indeed learned band-specific knowledge, we compare the sensitivity of B01 to water with that of other bands. In Fig. 7, we show the sensitivity of different bands of the GCBF method to water in the first two rows. The corresponding bands are labeled in the upper right corner of each figure, and the regions of water and urban fabric are marked with red dotted boxes. It can be seen that the sensitivity of band B01 to water is far superior to other bands.

Fig. 8 presents the visualization of the B8A band, where our method has learned that this band is sensitive to identifying urban fabric and can accurately detect it in complex terrain. We show the sensitivity of the B8A band to urban fabric in the last two rows of Fig. 7. Combining the results presented in Tables II and IV, it can be seen that our model has learned the unique knowledge of each band and exhibited strong multispectral image understanding and classification abilities by mining the complementary information between various bands.

TABLE V

RESULTS OF THE ABLATION STUDY ON BIGEARTHNET. SPECIFICALLY, IT INCLUDES THE NECESSITY OF USING MORE SPECTRAL BAND DATA AND THE EFFECTIVENESS OF BAND GROUPING, GROUPED PATCH EMBEDDING, CBAF, AND UNIFORMER BRANCH

more bands	band grouping	grouped patch embedding	CBAF (concat)	UniFormer branch	mAP
✗	✓	✓	✓	✓	87.7
✓	✗	✓	✓	✓	88.7
✓	✓	✗	✓	✓	89.2
✓	✓	✓	concat	✓	88.9
✓	✓	✓	channel attention	✓	89.1
✓	✓	✓	✓	✗	89.0
✓	✓	✓	✓	✓	**89.7**

E. Ablation Studies

1) Necessity of Using More Spectral Band Data: In our GCBF, we input RGB images into UniFormer and input other band data into the MGSFE network. Then, we fuse the outputs of both through the GCBF module to complete the final classification. In order to demonstrate the usefulness of utilizing information from other bands for our proposed method, we present the results of using only UniFormer to classify RGB bands in the first row of Table V. From the results, it is evident that our approach achieves superior classification performance by leveraging information from additional bands beyond RGB.

2) Significance of Band Grouping: Considering that each spectral band is able to provide unique reflection characteristics of the scene, we believe that if spectral bands are not grouped, the significant features of each band may be overwhelmed by information from other bands. Therefore,

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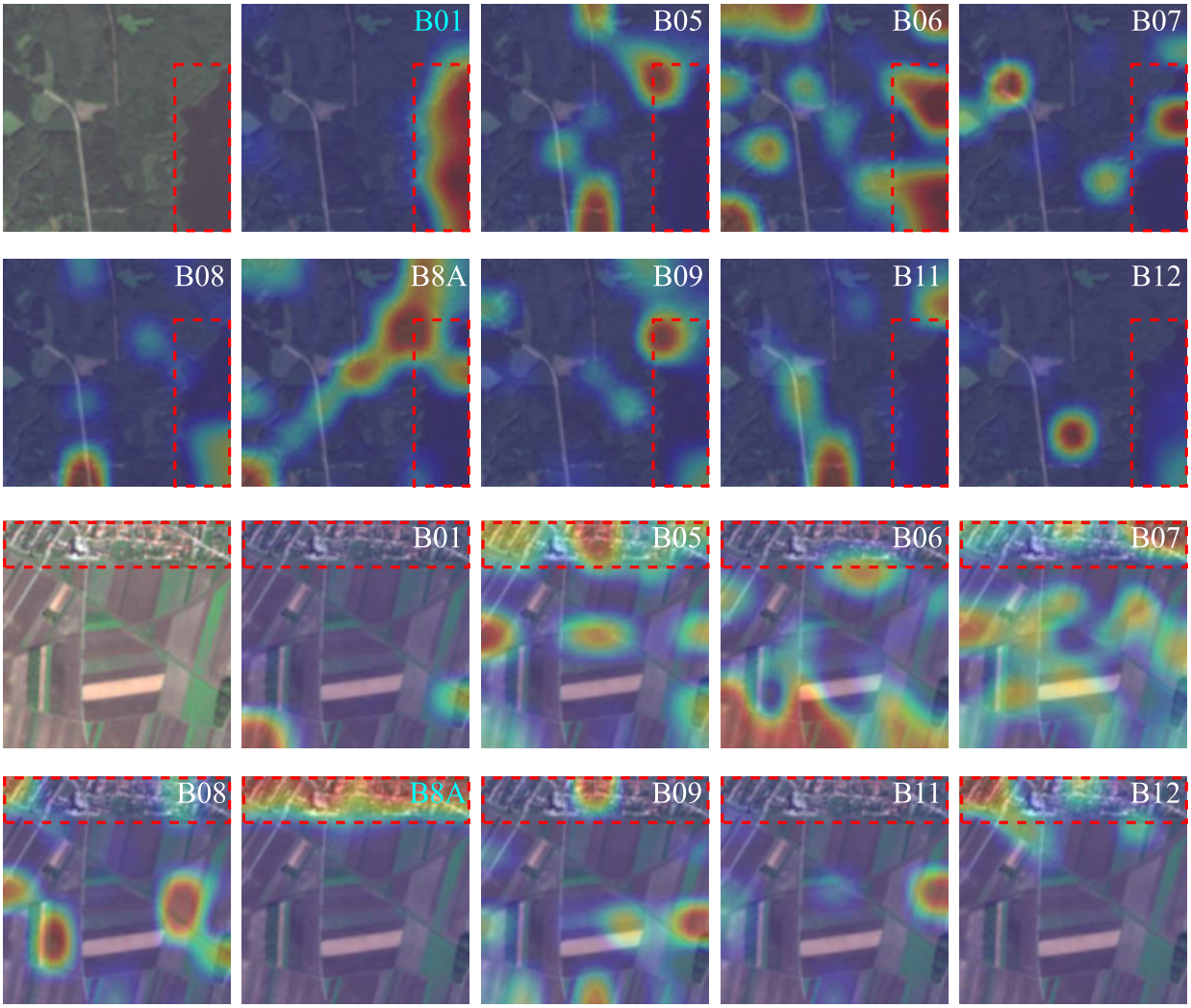


Fig. 7. Class activation maps of GCBF for all bands except B02, B03, and B04. The first two rows and the last two rows show the class activation maps for "water" and "urban fabric," respectively. The corresponding bands are labeled in the upper right corner of each figure, and the regions of water and urban fabric are marked with red dotted boxes. It can be seen that the proposed GCBF learned that the sensitivity of bands B01 and B8A to "water" and "urban fabric" is far superior to that of other bands.

to better extract and maintain the features of each band, an important operation that we perform is to group the bands in MGSFE. The classification results without band grouping in MGSFE are presented in the second row of Table V to verify the significance of band grouping. It can be observed that our proposed band grouping operation can further improve the classification performance for multispectral images.

3) Effectiveness of Grouped Patch Embedding Strategy: For visual models, in order to build a hierarchical feature pyramid, the feature maps are usually downsampled at the beginning of each stage. In our GCBF, to maintain the feature extracted from each band group during the downsampling stage without being affected by other bands, we use grouped patch embedding. To verify the effectiveness of our grouped patch embedding, we replace all the grouped patch embedding layers with normal patch embedding layers and report the quantitative results of this strategy in the third row of Table V. As expected, our grouped downsampling method outperforms the normal downsampling strategy.

TABLE VI

QUANTITATIVE RESULTS OF THE CLASSIFICATION PERFORMANCE OF CHANGING THE NUMBER OF BANDS IN THE INPUT IMAGE OF MGSFE BRANCH NETWORK

Input band	mAP	OP	OR	OF1	CP	CR	CF1
RGB + [B08]	84.0	79.8	68.0	73.4	65.9	52.1	56.5
RGB + [B01, B06, B8A]	87.0	80.6	73.7	77.0	70.2	61.9	64.5
RGB + [B01, B05, B07, B09, B12]	87.5	80.9	74.6	77.6	71.4	61.9	64.8
RGB + [B01, B05, B06, B07, B08, B09, B12]	88.1	83.7	77.6	80.5	74.9	66.7	69.4
All bands	**89.7**	**85.7**	**81.3**	**83.4**	**77.6**	**70.8**	**73.1**

4) Influence of CBAF: Inspired by the band combination method, we designed the CBAF module to enable the model to adaptively combine different band groups for different scene categories. We investigate the influence of CBAF by replacing the CBAF module with a simple concatenation. In addition, for a more comprehensive comparison, we also replace CBAF with standard channel attention architecture [44] for experimentation. The results in the fourth and fifth rows of Table V show that our CBAF is helpful in taking full advan-

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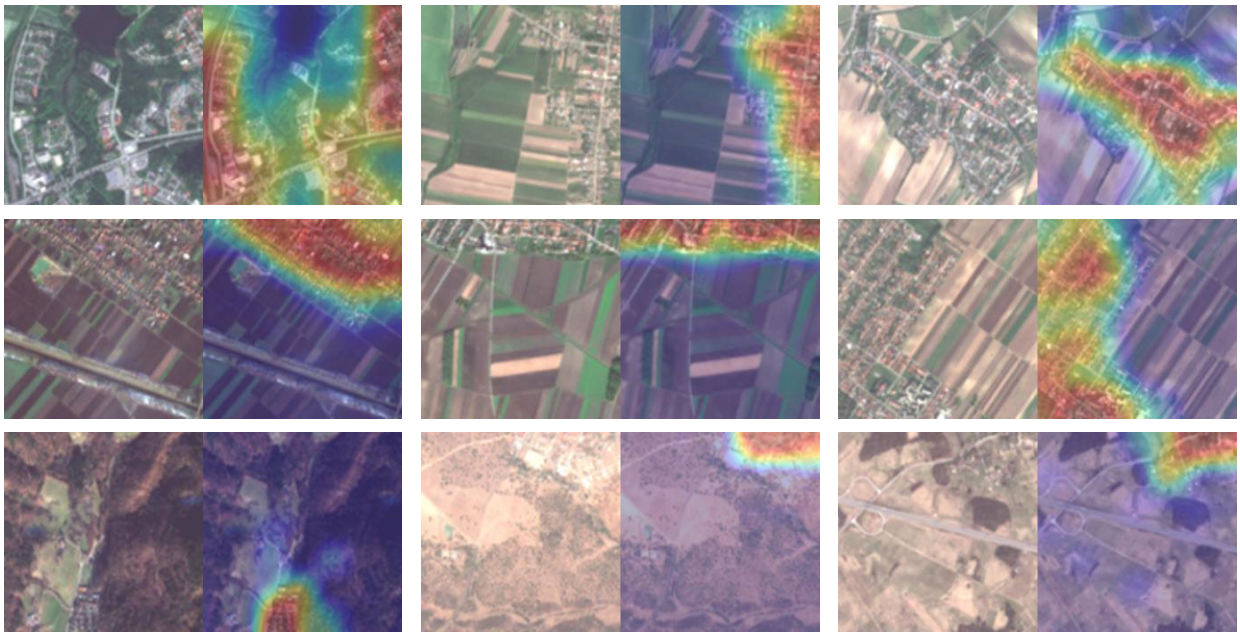


Fig. 8. Class activation maps for "urban fabric" in band B8A of GCBF. It is clear that band B8A is perceptive to the urban fabric contained in images, and our model has learned discriminative information in band B8A.

tage of complementary information between different spectral bands.

5) Necessity of UniFormer Branch for RGB: In our GCBF, we introduce the additional UniFormer branch to handle the RGB bands separately. This is because UniFormer was initially designed for processing RGB images, and using this network can better extract features from RGB images. To verify the necessity of adding the UniFormer branch, we present the performance of using only the MGSFE to process all bands in the sixth row of Table V. It is evident that using the additional UniFormer branch specifically to handle RGB bands is helpful for understanding multispectral images.

F. Impact of Free Setting

In order to fully demonstrate the capabilities and effectiveness of GCBF, we explore the impact of free setting on our method. Specifically, we explore the impact of different band numbers on the classification performance of our method on the BigEarthNet dataset by changing the number of bands in the images input to the MGSFE branch. The quantitative experimental results are presented in Table VI. It should be noted that the input bands are randomly selected. It can be seen that as the number of selected bands increases, the performance of our method in all metrics improves. This is because as the input band increases, our method can learn more complementary information from the image, thereby improving the classification performance of the model.

G. Model Complexity

In addition to the performance of the model, we are also concerned with the space cost of the model and the time cost during the inference process. We measure the space and time cost of the model using the number of parameters and FLOPs, respectively. In Table VII, we quantitatively compare GCBF with other methods in terms of the number of parameters and FLOPs. It can be seen that our method can achieve the best classification results with an appropriate number of parameters and computations, even outperforming the ViT-L, which has the largest number of parameters and FLOPs. In summary, our model achieves a good balance between performance and the efficiency of time and space.

TABLE VII
QUANTITATIVE COMPARISON RESULTS OF THE NUMBER OF PARAMETERS AND FLOPS BETWEEN OUR GCBF AND THE COMPARATIVE METHODS

Method	Parameters(M)	FLOPs(G)
ResNet-152	58.2	12.0
ViT-L	305.7	60.1
Swin-B	86.9	23.6
EMTCAL	27.8	4.6
GCSANet	8.2	3.3
EFPN-DSE-TDFF	25.7	9.8
BWMS	11.3	8.3
GCBF(ours)	119.4	19.3

V. CONCLUSION

In this article, we attempt to explore the remote sensing multispectral image scene classification and propose a new network called GCBF. In particular, we first group the various spectral bands, combining the RGB bands into one group and treating each of the other bands as an individual group. Subsequently, we utilize the existing UniFormer to extract the features from RGB bands. Meanwhile, we propose MGSFE to learn the distinct representations of each band group other than RGB. Finally, we are inspired

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by the band combination and design the CBAF module that can adaptively fuse the output of UniFormer and MGSFE. Compared with the existing methods for multispectral remote sensing image classification, the proposed GCBF can mine the complementary information between different band groups through the cross-attention mechanism, resulting in improved classification performance. Extensive experiments are built on three widely used multispectral datasets of BigEarthNet, SEN12MS, and EuroSAT that demonstrate the excellent multispectral image understanding capability of our GCBF.

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